
MATH 562: STATISTICAL INFERENCE

Homework 3

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1 Preliminaries

We hope to introduce the reader to the main ideas and methods that will be used throughout this work.

Definition 1.1 (Chain rule for densities). For random variables $Y_1, \dots, Y_n$,

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{Y_1}(y_1) \prod_{j=2}^n f_{Y_j|Y_1, \dots, Y_{j-1}}(y_j | y_1, \dots, y_{j-1}).$$

This identity, called the *prediction decomposition*, expresses the joint density as the product of one-step-ahead conditional densities.

1.1 AR(1) Gaussian Process

The following process is an autoregressive model of order 1 (AR(1)):

$$Y_j = \mu + \alpha(Y_{j-1} - \mu) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma^2),$$

which satisfies the following conditional distribution:

$$Y_j | Y_{j-1} = y_{j-1} \sim \mathcal{N}\{\mu + \alpha(y_{j-1} - \mu), \sigma^2\}.$$

1.2 Likelihood

As a preliminary note, we define the likelihood function $L(\mathbf{y}|\theta)$ as $\mathbb{P}(\text{seeing data}|\theta) = \mathbb{P}(y_1, \dots, y_n|\theta)$ which ‘splits’ due to the property of independence. More formally,

Definition 1.2 (Likelihood). Let $\mathbf{y} = (y_1, \dots, y_n)$ be i.i.d. samples from a probability mass function (pmf) $p_Y(t|\theta)$ (if Y is discrete) or probability density function (pdf) $f_Y(t|\theta)$ (if Y is continuous) parameterized (with parameter) by $\theta \in \Theta$. We define the likelihood of $\mathbf{y}$ given θ to be the “probability” of observing $\mathbf{y}$ if the true parameter is θ . If Y is discrete,

$$L(\mathbf{y}|\theta) = \prod_{i=1}^n p_Y(y_i|\theta)$$

and if Y is continuous,

$$L(\mathbf{y}|\theta) = \prod_{i=1}^n f_Y(y_i|\theta).$$

Hence, the likelihood function for data $y_0, \dots, y_n$ from an AR(1) Gaussian process with known initial value y_0 is given by

$$L(\mu, \alpha, \sigma^2) = \prod_{j=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} [y_j - \mu - \alpha(y_{j-1} - \mu)]^2 \right\}.$$

Remark 1.3. Note that the log-likelihood function is just the logarithm of the likelihood function. It is sometimes more convenient to work with since the logarithm of a product is the sum of the logarithms hence, it transforms products into sums. More formally, the log-likelihood function is given by

$$\ell(\mathbf{y}|\theta) := \log_e L(\mathbf{y}|\theta) = \sum_{i=1}^n \log_e f_Y(y_i|\theta)$$

For the discrete case, switch f_Y with p_Y in the above expression.

2 Exercise

2.1 Defining the problem

Observations $Y_1, \dots, Y_n$ arise in time order.

1. Starting from

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{Y_n|Y_1, \dots, Y_{n-1}}(y_n|y_1, \dots, y_{n-1})f_{Y_1, \dots, Y_{n-1}}(y_1, \dots, y_{n-1}),$$

establish the *prediction decomposition*

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{Y_1}(y_1) \prod_{j=2}^n f_{Y_j|Y_1, \dots, Y_{j-1}}(y_j|y_1, \dots, y_{j-1}).$$

2. A stationary first-order Gaussian autoregressive process satisfies

$$Y_j|Y_1 = y_1, \dots, Y_j|Y_{j-1} = y_{j-1} \sim \mathcal{N}(\mu + \alpha(y_{j-1} - \mu), \sigma^2) \text{ for } j = 1, \dots, n$$

with $|\alpha| < 1$, $\mu \in \mathbb{R}$, and $\sigma^2 > 0$. Find the log likelihood for data $y_0, \dots, y_n$ for this model if the initial value y_0 is treated

- (a) fixed as known constant,
 - (b) coming from the stationary distribution $\mathcal{N}(\mu, \frac{\sigma^2}{1-\alpha^2})$
3. Give a minimal sufficient statistic for $\theta = (\mu, \sigma^2, \alpha)$ in (2). Does the model with parameter vector θ form an exponential family?

2.2 Solution

We first show the prediction decomposition formula starting from the given identity on (1).

2.2.1 Prediction decomposition derivation

$$\begin{aligned} f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) &= f_{Y_n|Y_1, \dots, Y_{n-1}}(y_n|y_1, \dots, y_{n-1})f_{Y_1, \dots, Y_{n-1}}(y_1, \dots, y_{n-1}) \\ &= f_{Y_n|Y_1, \dots, Y_{n-1}}(y_n|y_1, \dots, y_{n-1})f_{Y_{n-1}|Y_1, \dots, Y_{n-2}}(y_{n-1}|y_1, \dots, y_{n-2})f_{Y_1, \dots, Y_{n-2}}(y_1, \dots, y_{n-2}) \\ &= f_{Y_n|Y_1, \dots, Y_{n-1}}(y_n|y_1, \dots, y_{n-1})f_{Y_{n-1}|Y_1, \dots, Y_{n-2}}(y_{n-1}|y_1, \dots, y_{n-2}) \cdots f_{Y_2|Y_1}(y_2|y_1)f_{Y_1}(y_1) \\ &= f_{Y_1}(y_1) \prod_{j=2}^n f_{Y_j|Y_1, \dots, Y_{j-1}}(y_j|y_1, \dots, y_{j-1}). \end{aligned}$$

This shows that the joint density equals the product of the onestep ahead predictive densities. In other words, the joint behaviour of the whole sequence is built by multiplying all the “predictive” pieces.

2.2.2 Likelihood functions for AR(1) process

Now, suppose we observe data $y_1, \dots, y_n$ from the autoregressive process

$$Y_j = \mu + \alpha(Y_{j-1} - \mu) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma^2),$$

with $|\alpha| < 1$. Note that the **gaussian** noises ε_j 's are i.i.d. This model means each Y_j depends linearly on the previous value Y_{j-1} plus some gaussian noise. Hence, this implies that

$$Y_j | Y_{j-1} \sim \mathcal{N}(\mu + \alpha(y_{j-1} - \mu), \sigma^2).$$

Case (i): Dealing with y_0 as a known constant. If the initial value y_0 is treated as a fixed, known constant, then we may write the joint pdf of the realizations $(y_1, \dots, y_n)$ of the observations as:

$$f(y_1, \dots, y_n | \mu, \alpha, \sigma^2) = \prod_{j=1}^n f(y_j | y_{j-1}; \mu, \alpha, \sigma^2)$$

And so, we might write the conditional log-likelihood as

$$\ell(\mu, \alpha, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n [y_j - \mu - \alpha(y_{j-1} - \mu)]^2.$$

Here y_0 enters only in the first term (i.e. the term $y_1 | y_0$), but no density is assigned to y_0 by itself.

Case (ii): Dealing with the devil; y_0 is random, from stationary distribution. If we assume the process is stationary, then

$$Y_0 \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{1 - \alpha^2}\right),$$

and so, we get that the marginal (stationary) distribution of the first observation is

$$Y_1 \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{1 - \alpha^2}\right).$$

Integrating out the unobserved Y_0 leads to the *exact likelihood*

$$L_e(\mu, \alpha, \sigma^2; y_1, \dots, y_n) = f\left(y_1; \mu, \frac{\sigma^2}{1 - \alpha^2}\right) \prod_{j=2}^n f(y_j | y_{j-1}; \mu, \alpha, \sigma^2),$$

and the corresponding log-likelihood is

$$\ell_e(\mu, \alpha, \sigma^2) = -\frac{1}{2} \log \frac{2\pi\sigma^2}{1 - \alpha^2} - \frac{(y_1 - \mu)^2(1 - \alpha^2)}{2\sigma^2} - \frac{n-1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=2}^n [y_j - \mu - \alpha(y_{j-1} - \mu)]^2.$$

Interpretation (for me to remember ...):

- The **conditional likelihood** treats y_0 as a fixed starting point and models only the transitions $Y_j | Y_{j-1}$ for $j = 1, \dots, n$.
- The **exact likelihood** assumes Y_0 (and hence Y_1) come from the stationary distribution and therefore includes their marginal densities.
- Both forms give consistent estimates of (μ, α, σ^2) as $n \rightarrow \infty$, but they differ slightly in small samples.

2.2.3 Minimal Sufficient Statistic

Note that likelihood depends on the data only through (if expanded and rearranged, the term in the summation that is)

$$\hat{T}(y) = \left(\sum_{j=1}^n y_j, \sum_{j=1}^n y_{j-1}, \sum_{j=1}^n y_j y_{j-1}, \sum_{j=1}^n y_j^2, \sum_{j=1}^n y_{j-1}^2 \right),$$

Remark 2.1. Note that the term $\sum_{j=1}^n y_{j-1}^2$ can be “omitted” (more like rewritten) since it can be computed from $\sum_{j=1}^n y_j^2$ and the known initial value y_0 (i.e. it is just lagged term of the y_j) as follows:

$$\sum_{j=1}^n y_{j-1}^2 = \sum_{j=0}^{n-1} y_j^2 = \sum_{j=1}^n y_j^2 + y_0^2 - y_n^2.$$

following the remark assbove, we define the following statistic:

$$T(y) = \left(\sum_{j=1}^n y_j, \sum_{j=1}^n y_{j-1}, \sum_{j=1}^n y_j y_{j-1}, \sum_{j=1}^n y_j^2 \right),$$

which is then minimal sufficient for $\theta = (\mu, \sigma^2, \alpha)$.

Exponential Family

Theorem 2.2 (Conditional AR(1) is an exponential family). Fix $y_0 \in \mathbb{R}$. Let $(Y_j)_{j=1}^n$ satisfy the Gaussian AR(1)

$$Y_j \mid Y_{j-1} = y_{j-1} \sim \mathcal{N}(\mu + \alpha(y_{j-1} - \mu), \sigma^2), \quad j = 1, \dots, n,$$

with parameter $\theta = (\mu, \alpha, \sigma^2)$ and $|\alpha| < 1$. Then the conditional joint density $f_\theta(y_1, \dots, y_n \mid y_0) = \prod_{j=1}^n f_\theta(y_j \mid y_{j-1})$ belongs to a (regular) exponential family.

Proof. Write the conditional density

$$f_\theta(y_j \mid y_{j-1}) = (2\pi\sigma^2)^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2} (y_j - \mu - \alpha(y_{j-1} - \mu))^2 \right\}.$$

Hence

$$\log f_\theta(y_{1:n} \mid y_0) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (y_j - \alpha y_{j-1} - (1-\alpha)\mu)^2.$$

Expanding the square and collecting terms that depend on the data gives

$$\log f_\theta(y_{1:n} \mid y_0) = C(\theta) + \eta_1(\theta) \sum_{j=1}^n y_j^2 + \eta_2(\theta) \sum_{j=1}^n y_{j-1}^2 + \eta_3(\theta) \sum_{j=1}^n y_j y_{j-1} + \eta_4(\theta) \sum_{j=1}^n y_j + \eta_5(\theta) \sum_{j=1}^n y_{j-1},$$

where the *natural parameters* $(\eta_k(\theta))$ are smooth functions of (μ, α, σ^2) (e.g. $\eta_1(\theta) = -\frac{1}{2\sigma^2}$, $\eta_3(\theta) = \frac{\alpha}{\sigma^2}$, etc.), and $C(\theta)$ *absorbs* all terms free of $y_{1:n} = (y_1, \dots, y_n)$. So

$$f_\theta(y_{1:n} \mid y_0) = h(y_{1:n} \mid y_0) \exp\left\{\sum_{k=1}^5 \eta_k(\theta) T_k(y_{1:n}) - A(\theta)\right\},$$

where $h(y_{1:n} \mid y_0) \equiv 1$ (I believe this function is sometimes called *carrier*), and sufficient statistics

$$T_1 = \sum y_j^2, \quad T_2 = \sum y_{j-1}^2, \quad T_3 = \sum y_j y_{j-1}, \quad T_4 = \sum y_j, \quad T_5 = \sum y_{j-1}.$$

This is exactly the canonical (normal, ‘bread and butter’) exponential-family form in $\mathbb{R}^5$, hence the claim. (Note that $-C(\theta) = A(\theta)$) $\square$

Remark 2.3 (On the role of Y_0). The statement is *conditional on* y_0 . If one assumes a stationary start $Y_0 \sim \mathcal{N}(\mu, \sigma^2/(1-\alpha^2))$, then the *exact* joint density $f_\theta(y_0, \dots, y_n)$ is also an exponential family after adding the marginal term for y_0 , with the same type of quadratic/linear sufficient statistics augmented by y_0 -terms. If Y_0 has unspecified law, only the conditional likelihood admits the exponential-family factorization.